DO NOW — RECALL

Lesson 148 · Network Hardware: Switches, Routers and

Name: _____ Class: _____ Date: _____

1. What does a firewall generally use to permit or block a connection?

2. What data structure follows FIFO?

3. What is the result of an AND gate with inputs 1 and 0?

4. What is a foreign key used for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A policy of rules, for example addresses, ports or connection state.

2. Model answer

A queue.

3. Model answer

0.

4. Model answer

Referencing a candidate key in another relation.

By the end, you can…

• Identify the role of switches, routers, wireless access points, NICs and repeaters.

• Follow a frame within a LAN and a packet between LANs, distinguishing MAC and IP use.

• Select network hardware for a realistic school network and justify its placement.

Vocabulary — know these terms

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

NEW LEARNING

Read, annotate and explain

Local connections

A switch forwards Ethernet frames using learned MAC addresses.

Across networks

Router consults IP destinations and routing information.

Wireless access

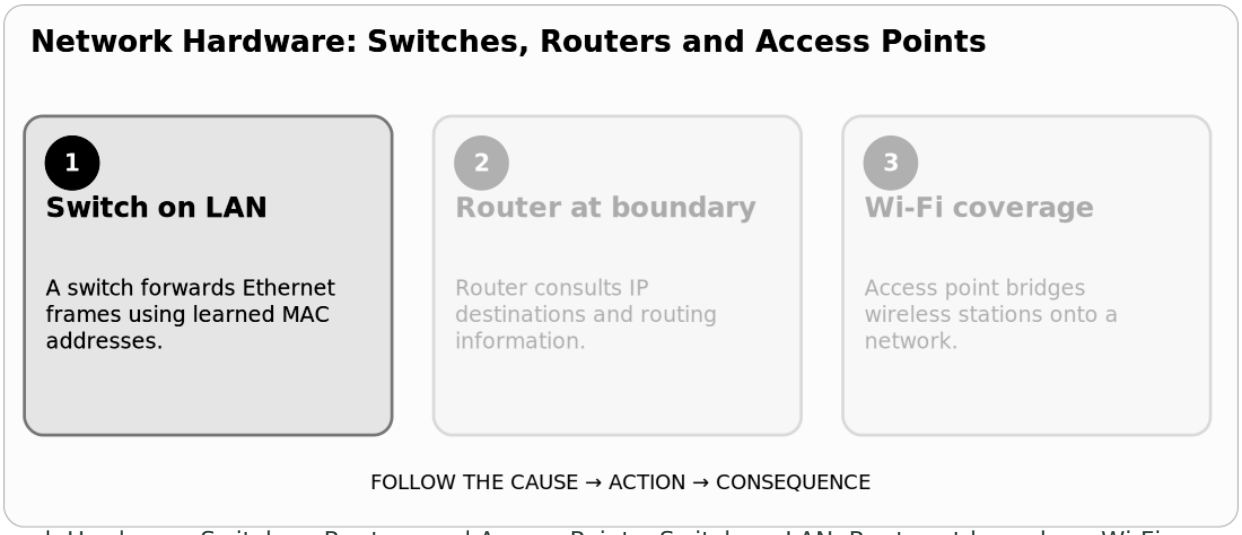
Access point bridges wireless stations onto a network.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

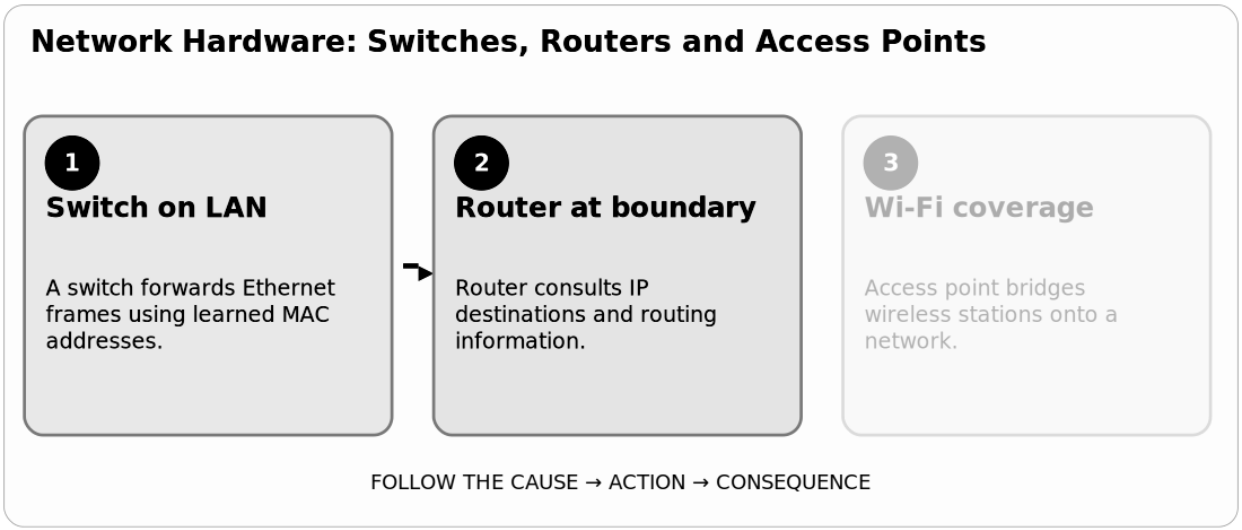
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Network Hardware: Switches, Routers and Access Points: Switch on LAN, Router at boundary, Wi-Fi coverage

New Learning

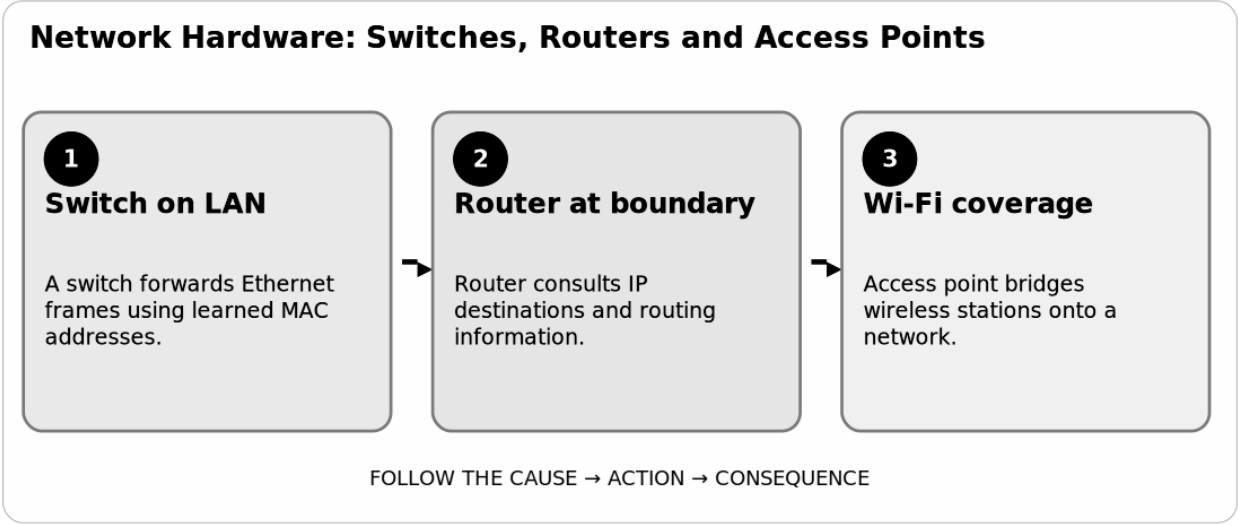


Network Hardware: Switches, Routers and Access Points: Switch on LAN, Router at boundary, Wi-Fi coverage

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New Learning



Network Hardware: Switches, Routers and Access Points: Switch on LAN, Router at boundary, Wi-Fi coverage

I DO — WORKED EXAMPLE

Follow the worked method

Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.

Laptop and switch

Identify the facts and make an initial prediction.

Router/default gateway

Explain the mechanism or trace the intermediate stage.

Remote network

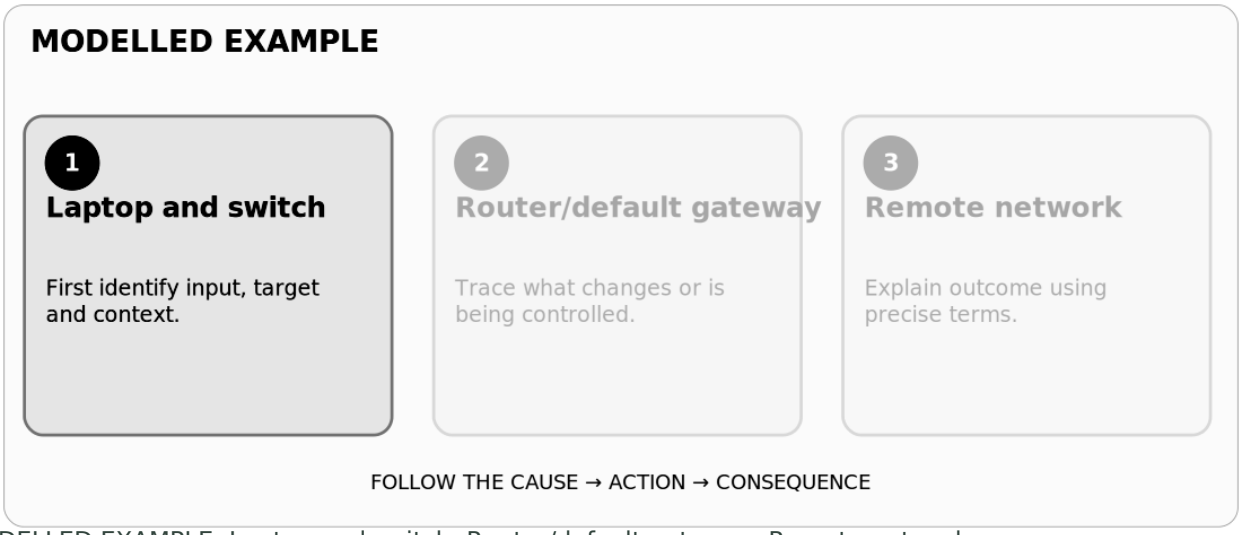
Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

I Do · original lesson diagrams

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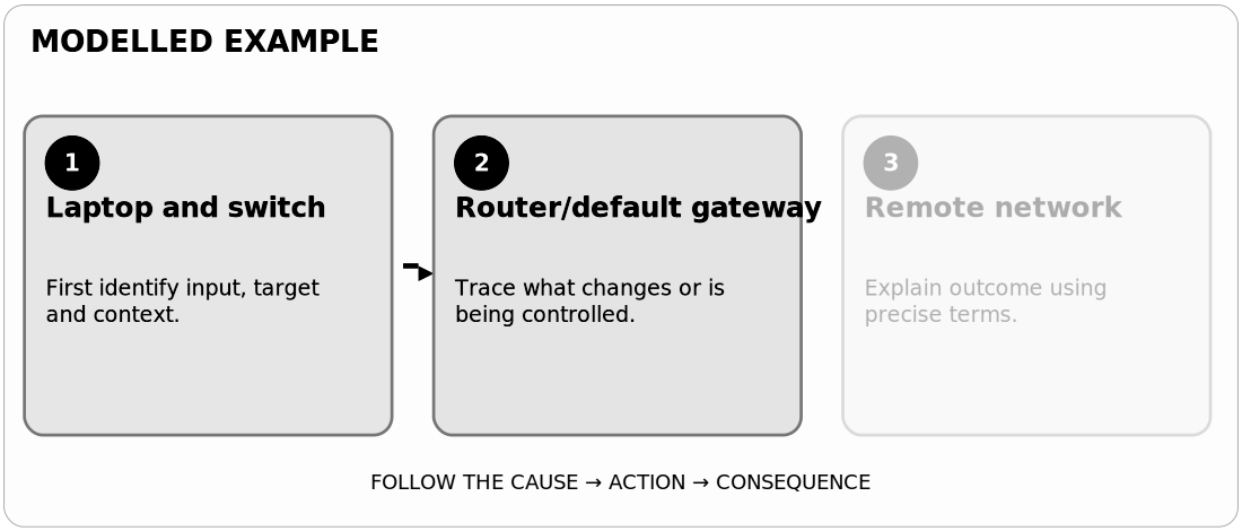
Identify the facts and make an initial prediction.



MODELLED EXAMPLE: Laptop and switch, Router/default gateway, Remote network

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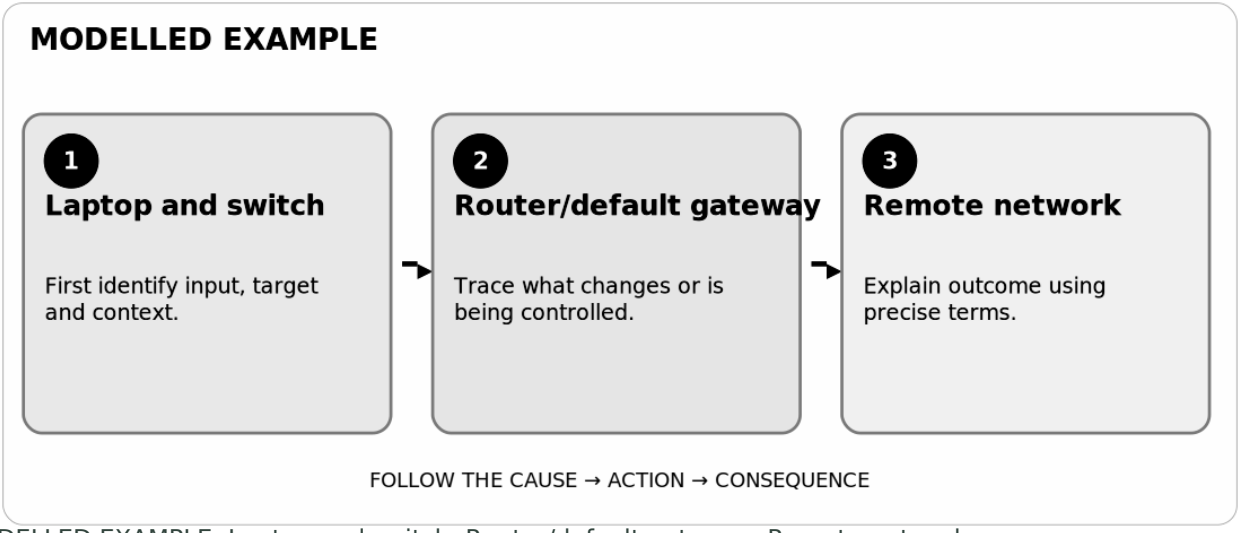
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WE DO — GUIDED PRACTICE

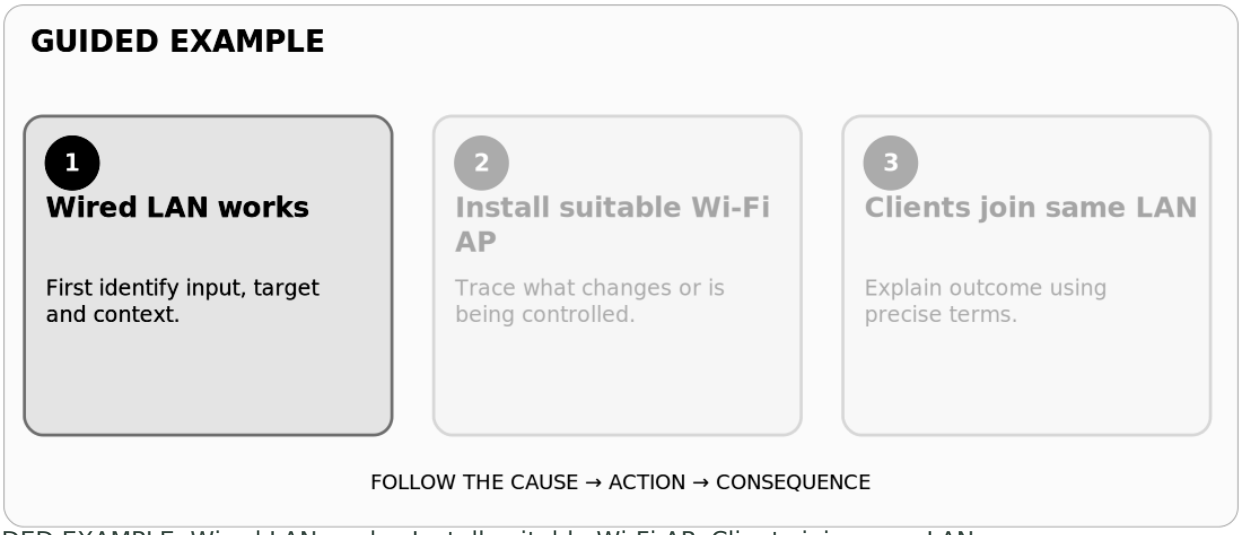
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A hall has strong wired connectivity but poor Wi-Fi. What hardware should be added, and why not another router

Identify the facts and make an initial prediction.



GUIDED EXAMPLE: Wired LAN works, Install suitable Wi-Fi AP, Clients join same LAN

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Wired LAN works Identify the facts and make an initial prediction.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A basic Ethernet switch chooses internet routes using destination website names.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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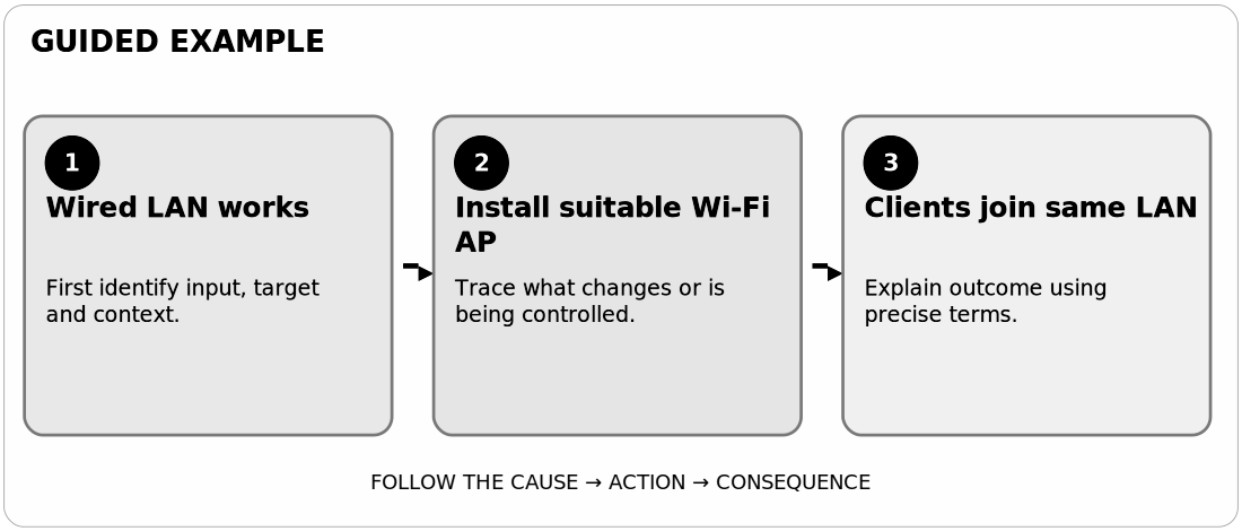
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GUIDED EXAMPLE: Wired LAN works, Install suitable Wi-Fi AP, Clients join same LAN

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GUIDED EXAMPLE: Wired LAN works, Install suitable Wi-Fi AP, Clients join same LAN

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Install suitable Wi-Fi AP

Explain the mechanism or trace the intermediate stage.

Clients join same LAN

Use a correctly connected wireless access point to extend radio coverage onto the existing LAN. An extra NAT router may create an unwanted separate network; plan channels, capacity and placement.

IMPROVE: Explain one correction to your own reasoning.